

Day 6: Introduction to Network Devices

Theory Assignment - Answers

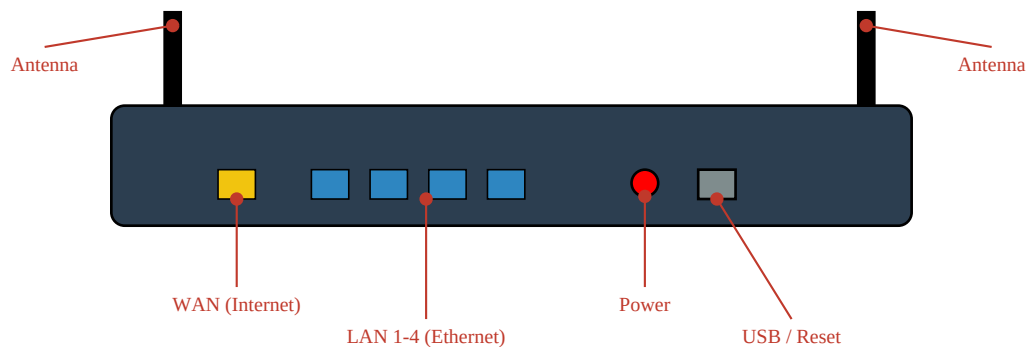
Q1. Functions and Typical Ports of Network Devices

Device	Main Function(s) in Home WiFi Setup	Typical Ports
Hub	Broadcasts every incoming signal to all connected devices (no intelligence). Rarely used today; causes collisions.	4-8 Ethernet (RJ-45) ports, power
Switch	Connects wired devices in the LAN; forwards frames only to the correct device using MAC addresses.	4-48 Ethernet (RJ-45) 10/100/1000, SFP uplink (optional), console (managed), power
Router	Connects home LAN to the Internet/ISP; assigns IPs (DHCP), does NAT and routing; usually has built-in switch + AP.	1 WAN/Internet port, 3-4 LAN Ethernet ports, power, USB (some), reset; antennas for WiFi
Firewall	Filters traffic by rules; blocks unauthorized access and threats between the LAN and Internet. Home routers have a built-in one.	WAN, LAN, DMZ (optional), console, power; filters logical ports e.g. 80 (HTTP), 443 (HTTPS), 53 (DNS)
Access Point (AP)	Provides wireless (WiFi) access to wired network; extends WiFi coverage.	1 Ethernet/PoE LAN port, power (DC or PoE), antennas (internal/external), console (some), reset

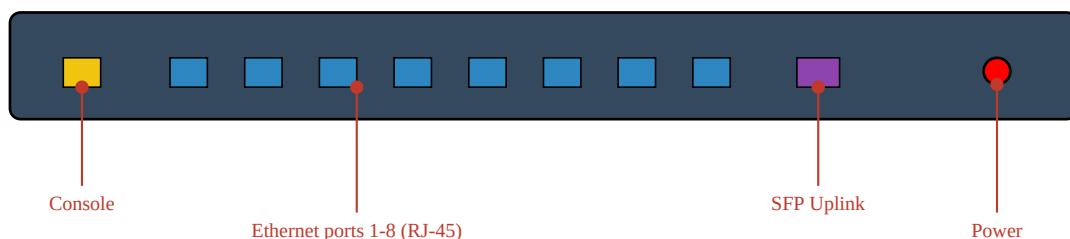
Q2. Labelled Device Diagrams with Ports

Since no images were attached, simplified diagrams of four devices are drawn below. Colour key: **blue** = Ethernet (LAN), **yellow** = WAN / Console, **purple** = SFP / Console, **red circle** = Power, black bars = Antennas.

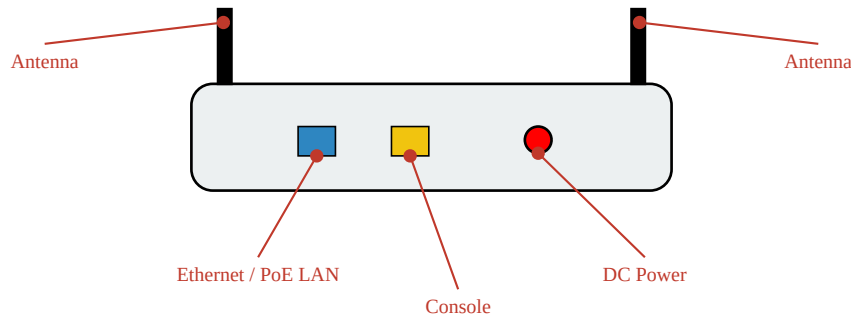
1. Wireless Router



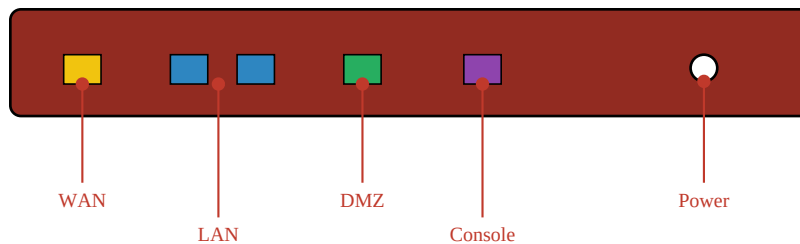
2. Network Switch (8-port, managed)



3. Wireless Access Point



4. Firewall Appliance



Q3. Role of Network Devices When Using Instagram

When I open Instagram on my phone at home, the **access point** (built into my WiFi router) receives my phone's wireless signal and connects it to the home network. The **router** then assigns my phone a private IP address, uses NAT to convert it to the public IP, and forwards my request through the ISP to Instagram's servers. The reply (photos and reels) comes back the same way: router, then access point, then my phone. The router's built-in **firewall** also blocks unwanted incoming traffic while allowing this response.

Q4. Slow and Disconnecting College WiFi

Possible reason: The **access points** are overloaded or too few. An older or low-capacity AP can handle only a limited number of simultaneous clients, so when hundreds of students connect, the airtime is shared, speeds fall, and devices get dropped. Weak coverage in some areas (classrooms far from an AP) also causes disconnections.

How upgrading helps: Replacing the old APs with modern **WiFi 6 (802.11ax) dual-band access points** and placing them properly gives higher client capacity (OFDMA, MU-MIMO), better coverage, and smoother handling of heavy traffic. Connecting them through a PoE switch with enough gigabit ports ensures the wired side does not become the next bottleneck.